Spreadsheet Formula Reference

A reference of the formulas used across the **Cleaned Data**, **Summary Tables**, and **Calculations** sheets. Descriptions assume column D is a quantity-type field, E a unit value, and F a total or amount; adjust if your headers differ.

1. Data Quality Checks

These run against the cleaned data range (rows 5–244).

Check	Formula	What it does
Blank cells	`=COUNTBLANK('Cleaned Data'!A5:H244)`	Counts empty cells across columns A–H. Should be 0.
Duplicate rows	`=SUMPRODUCT(--(COUNTIF('Cleaned Data'!A5:A244,'Cleaned Data'!A5:A244)>1))`	Counts rows whose column A value appears more than once. Should be 0.
Flagged rows	`=COUNTIF('Cleaned Data'!I5:I244,"Check")`	Counts rows flagged "Check" by the row-level test below. Should be 0.

Row-level validation (Cleaned Data, column I)

```
=IF(ABS(F5-D5*E5)<0.011,"OK","Check")
```

Flags a row as "Check" when F differs from D × E by more than about one cent (0.011 tolerance allows for rounding). Fill down from row 5 to row 244.

Note: the three data quality checks above appear a second time in your list. If both sets live on different sheets, keep them; otherwise one set can be removed.

2. Summary Metrics

Headline figures drawn from the cleaned data.

Metric	Formula
Total of F	`=SUM('Cleaned Data'!F5:F244)`
Record count	`=COUNTA('Cleaned Data'!A5:A244)`

Metric	Formula
Total of D	`=SUM('Cleaned Data'!D5:D244)`
Average of F	`=AVERAGE('Cleaned Data'!F5:F244)`
Average of D	`=AVERAGE('Cleaned Data'!D5:D244)`
Maximum of F	`=MAX('Cleaned Data'!F5:F244)`
Minimum of F	`=MIN('Cleaned Data'!F5:F244)`

3. Summary Table Row Counts

Confirms each summary table is fully populated.

Table	Formula	Rows covered
Table 1 (column A)	`=COUNTA('Summary Tables'!A6:A17)`	12
Table 2 (column F)	`=COUNTA('Summary Tables'!F6:F10)`	5
Table 3 (column K)	`=COUNTA('Summary Tables'!K6:K11)`	6

4. Share of Total

Each value is divided by a single grand total held in `Calculations!C5`. The `$` anchors keep the reference fixed when you fill down.

```
=D6/'Calculations'!$C$5
=I6/'Calculations'!$C$5
=N6/'Calculations'!$C$5
```

Format these cells as percentages. Each table's shares should sum to 100% if the tables cover the full total.

5. Growth Rate

```
=D25/D24-1
```

Period-over-period change: the current value (D25) relative to the prior value (D24). Format as a percentage.

Tips

- Wrap the share and growth formulas in `IFERROR(..., 0)` if the denominator could ever be zero.
- Replace fixed ranges like `A5:A244` with an Excel Table or named range so they expand automatically as data grows.
- The duplicate check uses `COUNTIF` over the whole range, which gets slow on very large datasets; it is fine at 240 rows.